TankGuard

Smart Water Tank Monitor — Know Exactly When to Clean Your Tank

Solar Powered · No Internet Needed · Phone Dashboard · Auto Pump Control

The Problem: India's Forgotten Water Tanks

Millions of homes, schools, hospitals, and apartment buildings store water in tanks. The water comes in clean — but sits for days or weeks. During that time:

What Happens Inside the Tank

- Chlorine (the disinfectant) disappears in 1–2 days
- Dirt and sludge settle at the bottom
- Green biofilm grows on walls and floor
- Bacteria and algae multiply

What You Experience

- Water smells or tastes bad — but it's too late
- Family gets stomach infections
- You waste water draining the whole tank to clean
- Cleaning is guesswork — every 6 months? Once a year?

> ***WHO says:*** *"The safe water chain breaks at the point of storage unless tanks are kept clean and covered." Today, nobody knows when to clean — until the water goes bad.*

20 Cr+
Household tanks in India

15 Lakh+
Institutional tanks (schools, hospitals, offices)

Zero
Affordable systems that tell you *when* to clean

100%
Current practice = guesswork

TankGuard's Solution: Guesswork → Evidence

A small solar-powered device sits on your tank lid. Sensors hang in the water and measure 5 things that actually change while water sits:

What It Measures	Why It Matters (Simple Terms)
Turbidity (cloudiness)	First sign of sludge & biofilm — rises before you can see it
pH (acid/alkaline balance)	Drift means algae, decay, or pipe corrosion
TDS (dissolved solids)	Minerals leaching from settled dirt and scale
Temperature	Warmer water = faster bacteria growth
Water Level	How long since last refill (water age)

These 5 numbers feed **TankScore** — one simple score that answers:

"Is it time to clean this tank?"

Answer: ✅ GOOD · ⚠️ WATCH · 🔴 CLEAN NOW

What Changes for You

- **Before:** Clean every 6 months (waste effort) or wait for bad smell (too late)
- **After:** Clean *exactly when needed* — get an alert on your phone
- **Before:** Climb the tank, open the lid, guess
- **After:** Open phone → see TankScore → pump runs only when needed
- **No WiFi? No internet? No problem.** Works completely offline. Your data never leaves your home.

What You Get & What It Costs

Core Kit (M1) — The Monitor

Everything needed to monitor one tank. Installs in 30 minutes.

Part	What It Does	Qty	Cost
Brain (ESP32 Dev Board)	Runs the system, connects to your phone via WiFi	1	₹1,237
Precision Converter (ADS1115)	Reads sensors accurately	1	₹145
pH Sensor Kit (Gravity SEN0161-V2)	Measures acid/alkaline balance	1	₹2,669
TDS Sensor (Gravity SEN0244)	Measures dissolved minerals	1	₹1,489
Turbidity Sensor (Gravity SEN0189)	Measures cloudiness (sludge indicator)	1	₹600
Temperature Probe (DS18B20 waterproof)	Water temperature	1	₹44
Water Level Sensor (JSN-SR04T ultrasonic)	Ultrasonic — measures fill level	1	₹270
Solar Panel (10W, 12V poly)	Powers everything from sunlight	1	₹950
Battery Pack (4×18650 Li-ion)	Runs 3 weeks without sun	1	₹760
Solar Charge Controller (CN3791 MPPT)	Manages battery charging	1	₹420
Weatherproof Box + Heat Shield (IP66)	Survives Indian summer, heat protected	1	₹1,520
Power Parts (buck, LDO, MOSFETs, wiring)	Safe 12V→5V→3.3V power rails	1	₹227
Calibration Kits & Spares	pH buffers, turbidity standards, probe care	1	₹2,060
Engineering & Integration	Assembly, calibration, bench validation	1	₹2,709
Core Kit Total			**₹15,500**

Core Monitoring Kit — ₹15,500

Auto Pump Control Module (M2) — Safe Pump Automation

NEW: A home-side unit next to the motor switch shows live tank status and switches the pump through an **opto-relay + motor-rated contactor** — the ESP8266 never touches mains. One button overrides the pump; a crash or lost link forces the pump **OFF**.

Part	What It Does	Qty	Cost
Opto-isolated Relay Module	5 V, 10 A / 250 V, isolates the ESP8266 from mains	1	**₹180**
Motor-rated Contactor	AC-3, 9 A, switches the pump safely	1	₹450
1.8″ ST7735 TFT	Live level, TankScore, pump/mode status	1	₹300
Enclosure, terminals, MCB	Separate LV/mains compartments, fusing	1	₹150
Pump Control Module Total			**₹1,080**

Auto Pump Control Module — ₹1,080

Total Project Cost (Parts Only)

Module	Cost
Core Monitor (M1)	₹15,500
Auto Pump Control (M2)	₹1,080
Total Parts Cost	**₹16,580**

Complete System Pricing — Parts Only

> **Note:** ₹16,580 is parts-only (excludes GST, freight, installation). The pump module is deliberately isolated: opto-relay → contactor → pump, with fail-safe OFF on crash or link loss. Mains wiring by a licensed electrician.

How We'll Build It in 2 Weeks

14 days. Daily milestones. Working system on a real tank by Day 14.

Week	Days	What We Do	You'll See
Week 1	1–2	Buy all parts locally; test each sensor on the bench	5 sensors showing live readings
	3–4	Calibrate each sensor with standard solutions (pH 4/7/9, TDS, turbidity)	Each sensor accurate within 2%
	5	Build solar+battery power; program deep-sleep (wakes every 3 hours)	Runs 48+ hours on battery alone
Week 2	6–7	Build TankScore brain: GOOD / WATCH / CLEAN NOW logic + trend detection	Brain correctly flags test water samples
	8–9	Build phone dashboard: gauges, charts, pump status, "Tank Cleaned" button	Open `tankmon.local` on phone — works offline
	10	Mount in weatherproof box; install on real tank; wire relay → contactor	Fully installed, solar panel angled
	11–13	Real-world test: 3-hour readings, watch TankScore change, test pump start/stop	72 hrs data; pump control works; score matches reality
14	Final report, user guide, live demo	**Working system ready to use**	

14-Day Build Plan

What Could Go Wrong (And How We Handle It)

- **Parts delayed?** Local vendors (Robu, Semicon, Amazon same-day) + backup suppliers identified
- **Sensors drift?** Calibrated Day 3–4; monitored daily during 72-hr test
- **Not enough sun?** Battery sized for 3 weeks without sun; tested Day 5
- **WiFi weak at tank?** Device creates its own hotspot as fallback; data never lost (stored locally)
- **Pump safety?** Opto-relay + contactor; pump forced OFF on crash or lost link

In 14 days you get: A solar-powered tank monitor + a safe home pump controller = **₹16,580 parts**. Phone dashboard works offline. Tells you *exactly when the water needs attention* and runs the pump only when needed. No cloud. No subscription. No guesswork.

Prepared: October 2026 · For project submission · Prices indicative (Indian market), subject to vendor confirmation